

MD. MANIRUZZAMAN

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EDUCATION

San Francisco Bay University

Fremont, CA, USA

Master of Science in Electrical Engineering

December 2025

- Cumulative GPA: **3.88** /4.00
- Honors: Recipient of the 100% Presidential Scholarship, President's List award for Spring 2025.
Recipient President's list award for the spring 2025 semester for GPA 4.0 out of 4.0.
- Relevant Coursework: Internet of Things (IoT), Data Science, Python Programming, Machine Learning, Blockchain, Quantum Computing, Computer Architecture, Digital Design HDL (Verilog).

North South University

Dhaka, Bangladesh

Master of Science in Electrical and Electronic Engineering

2016 - 2017

- Cumulative GPA: **3.37** /4.00
- Honors: Recipient of the 50% Dean Scholarship
- Relevant Coursework: RF and Microwave Engineering, Digital Communication Systems, Network Architecture and protocols, Power Electronics system Integration.

Green University of Bangladesh

Dhaka, Bangladesh

Bachelor of Science in Electrical and Electronic Engineering

2012 - 2016

- Cumulative GPA: **3.92** /4.00
- Honors: VC awards for 3 consecutive semesters and Dean Scholarship
- Relevant Coursework: Computer programming, Power Electronics, DC circuit, AC circuit, Transmission and Distribution, Electrical Motor, VLSI design, Analogue Integrated Circuit, Control System Engineering, Communication Theory, Digital Electronics, Digital Signal Processing, RF and Microwave Engineering, Microprocessor and Interfacing.

SKILLS

Programming Languages: Python, MIPS assembly language, Verilog HDL.

Python Libraries & Frameworks: NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Matplotlib, Seaborn and more.

Research & Technical skills: Data analysis, Machine Learning, Deep Learning, Artificial Intelligence (AI), Internet of Things (IoT), Blockchain-enabled systems.

Broadcast Skills (Radio Station): FM signal broadcasting, Raspberry Pi, FM Transmitter, Radio Studio, Radio Traffic, Radio Automation system, Audio Streaming, Music and Traffic scheduling, Audio Transmission Equipment installation.

PUBLICATION

14. Ghosh, H., Rahat, I.S., Hossain, M.Z. et al. TumorSageNet CNN hybrid architecture enables accurate detection of mango leaf pathologies. Sci Rep (2026). <https://doi.org/10.1038/s41598-026-40944-2>
13. Md Shakil Islam, Md. Maniruzzaman, & Md Abul Kalam Azad. (2025). An IoT-Based Remote Patient Monitoring Architecture Integrating Edge AI and Blockchain for Chronic Disease Management. Journal of Computer Science and Technology Studies, 7(11), 442-451. <https://doi.org/10.32996/jcsts.2025.7.11.41>
12. M. S. Islam, M. M. Hosen, M. B. Uddin, M. Hossain, M. Maniruzzaman and A. Akhtar, "MB-CNN-LSTM-ATTN: An Optimized Deep Learning Model for Sleep Disorder Classification," 2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS), Kushtia, Bangladesh, 2025, pp. 1-6, doi: 10.1109/COMPAS67506.2025.11381655.
11. Md. Maniruzzaman, Md. Imran Chowdhury Rana, Md. Firoz Kabir, Hritwik Ghosh, *et al.*, "Integrating Deep Learning in Prostate Cancer Grading: Innovations in Computational Pathology," in *Artificial Intelligence in Oncology*, Springer, Cham, Oct. 2025, ch. 27. doi: 10.1007/978-3-031-94302-7_27.
10. Maniruzzaman, M., Jaman, M. S., Abid, M. A. S., Mahmud, Z., Rahman, M. E., Siddiky, M. N. A., & Mahmud, Z. (2025, February). A hybrid mRMR RFE and AI framework for advancing Alzheimer's biomarkers discovery. In *Proceedings of the International Conference on Artificial Intelligence in Information and Communication (ICAIIIC 2025)*. <https://doi.org/10.1109/ICAIIIC64266.2025.10920700>.

09. A. Ghosh, M. F. I. Buian, M. Maniruzzaman, M. M. Hossain, A. K. Azad, A. A. Miazee, I. Ragab, A. A. Hassan, H. A. Alrafai, and S. K. A. Abdelrahim, "Numerical analysis and device modelling of a lead-free $\text{Sr}_3\text{PI}_3/\text{Sr}_3\text{SbI}_3$ double absorber solar cell for enhanced efficiency," *RSC Adv.*, vol. 14, pp. 26437–26456, 2024, doi: [10.1039/D4RA05079G](https://doi.org/10.1039/D4RA05079G).
08. Hassan, A. A., Ria, M. S. I., Ghosh, A., Alrafai, H. A., Baki, A. A., Abdelrahim, S. K. A., Al-Humaidi, J. Y., Robin, R. I. C., Rahman, M. M., & Maniruzzaman, M. (2024). Investigating the physical characteristics and photovoltaic performance of inorganic Ba_3NCl_3 perovskite utilizing DFT and SCAPS-1D simulations. *Materials Science and Engineering: B*, 2024, 117559. <https://doi.org/10.1016/j.mseb.2024.117559>.
07. A. Ghosh, M. S. Hossain, F. Ahmed, M. M. J. Juhi, M. M. Rahman, N. S. Awwad, H. A. Ibrahim, R. Kabir, M. F. Ahmmed, M. Maniruzzaman, and M. M. R. Sonic, "Improving the power conversion efficiency of RbPbBr_3 absorber-based solar cells through the variation of efficient hole transport layers," *J. Phys. Chem. Solids*, 2024, Art. no. 112179, doi: [10.1016/j.jpcs.2024.112179](https://doi.org/10.1016/j.jpcs.2024.112179).
06. M. Maniruzzaman, A. Sami, R. Hoque, and P. Mandal, "Pneumonia prediction using deep learning in chest X-ray images," *Int. J. Sci. Res. Arch.*, vol. 12, no. 1, pp. 767–773, 2024, doi: [10.30574/ijrsra.2024.12.1.0880](https://doi.org/10.30574/ijrsra.2024.12.1.0880).
05. R. Hoque, M. Maniruzzaman, D. L. Michael, and M. Hoque, "Empowering blockchain with SmartNIC: Enhancing performance, security, and scalability," *World J. Adv. Res. Rev.*, vol. 22, no. 1, pp. 151–162, 2024, doi: [10.30574/wjarr.2024.22.1.1026](https://doi.org/10.30574/wjarr.2024.22.1.1026).
04. A. Ghosh, M. J. Ferdous, M. M. J. Juhi, M. F. I. Buian, A. A. Miazee, M. Sajid, M. Maniruzzaman, A. M. Tighezza, M. F. Ahmmed, and M. S. Islam, "Strain-induced changes in the electronic, optical, and mechanical properties of the inorganic cubic halide perovskite Sr_3PBr_3 with FP-DFT," *J. Phys. Chem. Solids*, 2024, Art. no. 112053, doi: [10.1016/j.jpcs.2024.112053](https://doi.org/10.1016/j.jpcs.2024.112053).
03. E. Bazgir, E. Haque, M. Maniruzzaman, and R. Hoque, "Skin cancer classification using Inception Network," *World J. Adv. Res. Rev.*, vol. 21, no. 2, pp. 839–849, 2024, doi: [10.30574/wjarr.2024.21.2.0500](https://doi.org/10.30574/wjarr.2024.21.2.0500).
02. M. A. Rahman, E. Bazgir, S. M. S. Hossain, and M. Maniruzzaman, "Skin cancer classification using NASNet," *Int. J. Sci. Res. Arch.*, vol. 11, no. 1, pp. 775–785, 2024, doi: [10.30574/ijrsra.2024.11.1.0106](https://doi.org/10.30574/ijrsra.2024.11.1.0106).
01. M. Maniruzzaman, M. S. Uddin, M. B. Hossain, and K. Hoque, "Understanding COVID-19 through tweets using machine learning: A visualization of trends and conversations," *Eur. J. Adv. Eng. Technol.*, vol. 10, no. 5, pp. 108–114, 2023, doi: [10.5281/zenodo.13832057](https://doi.org/10.5281/zenodo.13832057).

PROJECT & RESEARCH EXPERIENCE

San Francisco Bay University

Fremont, CA

AI-based Resume analyser with recommended job and professional certification course (Capstone Project).

Fall 2025

- Introduce an AI-based resume analysis system to reduce the gap between students and jobs.
- The system will recommend a professional certification course to increase skills according to educational background.
- Also, recommend a job for which the existing resume is suitable (in terms of education, work experience, and skills).

San Francisco Bay University

Fremont, CA

Integrate IoT-based health monitoring systems with Tiny-ML and Blockchain

Summer 2025

- Introduce a new 6-layer architecture to overcome the previous study gap.
- Comes with Edge AI and a cloud server for quick prediction and remote access.

STANDARDIZED TEST SCORES

GRE (Graduate Record Examination)

Dhaka, Bangladesh

Registration Number: 3127318

29th October 2024

- Verbal: 166 (96th Percentile) - Quantitative: 159 (50th Percentile) - Analytical Writing: 3.0 (16th Percentile)

IELTS

Dhaka, Bangladesh

Candidate Number: 044634

17th December 2022

- Listening: 6.5 - Reading: 6.0 - Writing: 6.0 - Speaking: 6.5 - **Overall Band Score: 6.5**

PROFESSIONAL EXPERIENCE

Global Star Communication

Dhaka, Bangladesh

Position: Manager, Department of Technical

16th February 2021 – 20th December 2023

- Commercial and community FM radio station's equipment installation, design and integration as per design.
- FM transmitter and broadcast equipment troubleshooting and repair. Radio automation system installation.

Radio Masala Limited (96.4 Spice FM)

Dhaka, Bangladesh

Position: Manager, Department of Technical

1st April 2019 – 15th February 2021

- Manage the technical team with a weekly roster of the team and work distribution among the team.
- Maintenance of Broadcast equipment (EU and USA), studio equipment (EU), and Radio automation software (ENCO).

Radio Masala Limited (96.4 Spice FM)

Dhaka, Bangladesh

Position: Sr. Broadcast and Telecom Manager

1st September 2016 – 31st March 2019

- Transmitter, Transmission and studio equipment maintenance and repair.
- Radio Automation software and server maintenance and troubleshooting.

LEADERSHIP

Second Harvest of Silicon Valley

San Jose, CA

Volunteer

3-hours

- Serve food to 100+ homeless. Prepare food, set up tables, and greet attendees.

Sacred Heart Community Service

Santa Clara, CA

Volunteer

2-hours

- Serve food to 50+ homeless. Prepare food, set up tables and serve them.